

VAN DINH NGHIA

FPGA & AI CHIP PORTFOLIO

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PROFILE

Electronics and Telecommunications Engineering student focused on Verilog RTL, FPGA implementation, SoC integration, and quantized AI acceleration. I take designs from architecture and bit-exact verification through timing closure, embedded software integration, and physical-board evaluation.

SELECTED PROJECT 01

INT8 1D-CNN AI Inference Accelerator

01/2026 – 08/2026

FPGA/SoC AI Hardware Designer — Xilinx KV260

98.31% Board accuracy	15,009 Samples evaluated	100 MHz Timing met	0 / 0 DSP / BRAM
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PROBLEM & APPROACH

The project deploys a five-class 1D-CNN on an FPGA/SoC platform. Floating-point feature extraction was converted to signed INT8 computation using per-layer scale and zero-point parameters, fixed-point requantization, rounding, saturation, and integer-only convolution, ReLU, and max-pooling.



ENGINEERING CONTRIBUTIONS

- Designed an accelerator with **20 MAC lanes**, **5 requantization units**, **42 execution contexts**, configurable on-chip memories, status/error reporting, and AXI4-Lite control.
- Partitioned inference between the FPGA feature extractor and the ARM processor, which executes the GAP/Dense classification tail.
- Developed a Linux ARM64 C application for parameter loading, inference control, output readback, accuracy measurement, and confusion-matrix generation.
- Built unit tests, a bit-accurate integer golden model, full-model RTL tests, and end-to-end physical-board validation.

MEASURED RESULTS

RTL/golden validation	10,240/10,240 outputs matched across 8 samples; predictions matched the quantized TFLite model.
Timing and latency	+0.701 ns WNS at 100 MHz; 21,437 cycles per inference.
Placed utilization	17,530 LUTs (14.97%), 6,898 registers (2.94%), 0 DSP, and 0 BRAM.
Physical-board evaluation	14,756/15,009 samples classified correctly on the KV260: 98.31% accuracy .

SELECTED PROJECT 02

FPGA Multiplier Architecture Optimization

10/2025 – 03/2026

Principal Investigator — Student Scientific Research Project

9 Architectures	114/114 Tests passed	725.16 MHz Best Fmax	3 Boards verified
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OBJECTIVE

Design and experimentally compare signed 16-bit FPGA multiplier architectures across functionality, area, maximum frequency, latency, throughput, and estimated power, then validate selected designs on multiple physical boards.

ARCHITECTURES IMPLEMENTED

- DSP baseline
- 2-stage pipeline
- Booth radix-4
- Array
- 8-stage pipeline
- 2-core Booth radix-4
- Shift-and-add
- Booth radix-2
- 8-core Booth radix-4

ENGINEERING CONTRIBUTIONS

- Led architecture selection, Verilog RTL development, reusable testbench creation, synthesis, implementation, metric comparison, board deployment, and final technical reporting.
- Collected LUT, FF, DSP, Fmax, latency, throughput, and power results to expose the trade-offs among combinational, sequential, pipelined, and parallel architectures.
- Designed a 115,200-baud UART hardware-validation flow with automated C-based PASS/FAIL checking.
- Deployed and verified selected designs on **Xilinx KV260**, **ZedBoard**, and **Tang Nano** hardware.

SELECTED RESULTS

Functional verification	All 114/114 test cases passed.
Maximum frequency	8-stage pipeline: 725.16 MHz ; array baseline: 108.57 MHz — nearly 7× higher Fmax .
Sequential latency	Booth radix-4 completed in 8 cycles , half the 16-cycle radix-2 latency.
Parallel throughput	8-core architecture completed eight results every eight cycles, sustaining one result/cycle .
Hardware validation	Stable host-to-board checking over UART at 115,200 baud on three FPGA platforms.

KEY TAKEAWAYS

- Deep pipelining maximized clock frequency but increased latency and register usage; the 2-stage design provided a more balanced frequency/latency trade-off.
- Booth radix-4 was a practical sequential option, while multi-core designs exchanged area and dynamic power for higher throughput.
- Physical-board testing revealed interface and integration concerns that were not visible in RTL simulation alone.